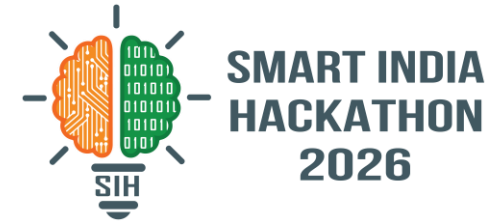


SMART INDIA HACKATHON 2026



- Problem Statement ID – Open Innovation
- Problem Statement Title- Crowdsourced Civic Issue Reporting and Resolution System
- Theme- Clean and green technology
- PS Category- Software
- Team ID- NA
- Team Name- Team Eka



CiviConnect: AI-Powered Crowdsourced Civic Issue Reporting & Resolution System

PROBLEM

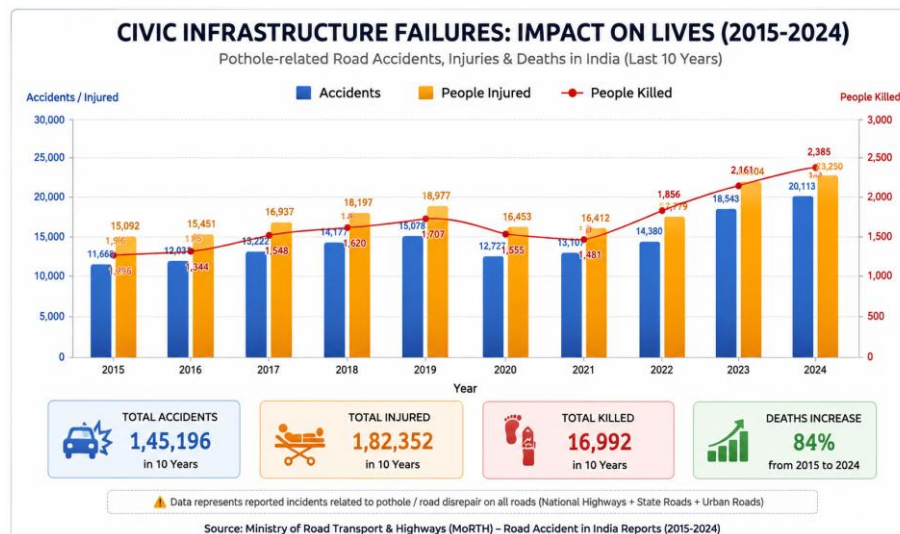
India's cities face thousands of unresolved civic issues such as potholes, garbage accumulation, broken streetlights, waterlogging and damaged public infrastructure. Fragmented reporting channels, manual complaint handling, incorrect department routing and lack of real-time tracking often lead to delayed resolutions, repeated complaints and increased public-safety risks.

SOLUTION

Our idea is to create an AI-powered Crowdsourced Civic Issue Reporting & Resolution Platform that connects citizens with the appropriate authorities through intelligent reporting, prioritization and real-time tracking, enabling faster, transparent and accountable civic issue resolution.

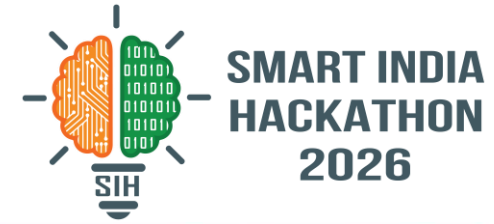
- AI-powered civic issue detection & classification
- GPS-based reporting with photo/video evidence
- Smart priority scoring & automatic department routing
- Real-time tracking, resolution verification & civic analytics

“From citizen reporting to verified resolution one intelligent civic ecosystem”.



TEAM
EKA

TECHNICAL APPROACH



Technologies to be used:

Backend Development- Java, Spring Boot, Spring Security (JWT), Spring data JPA

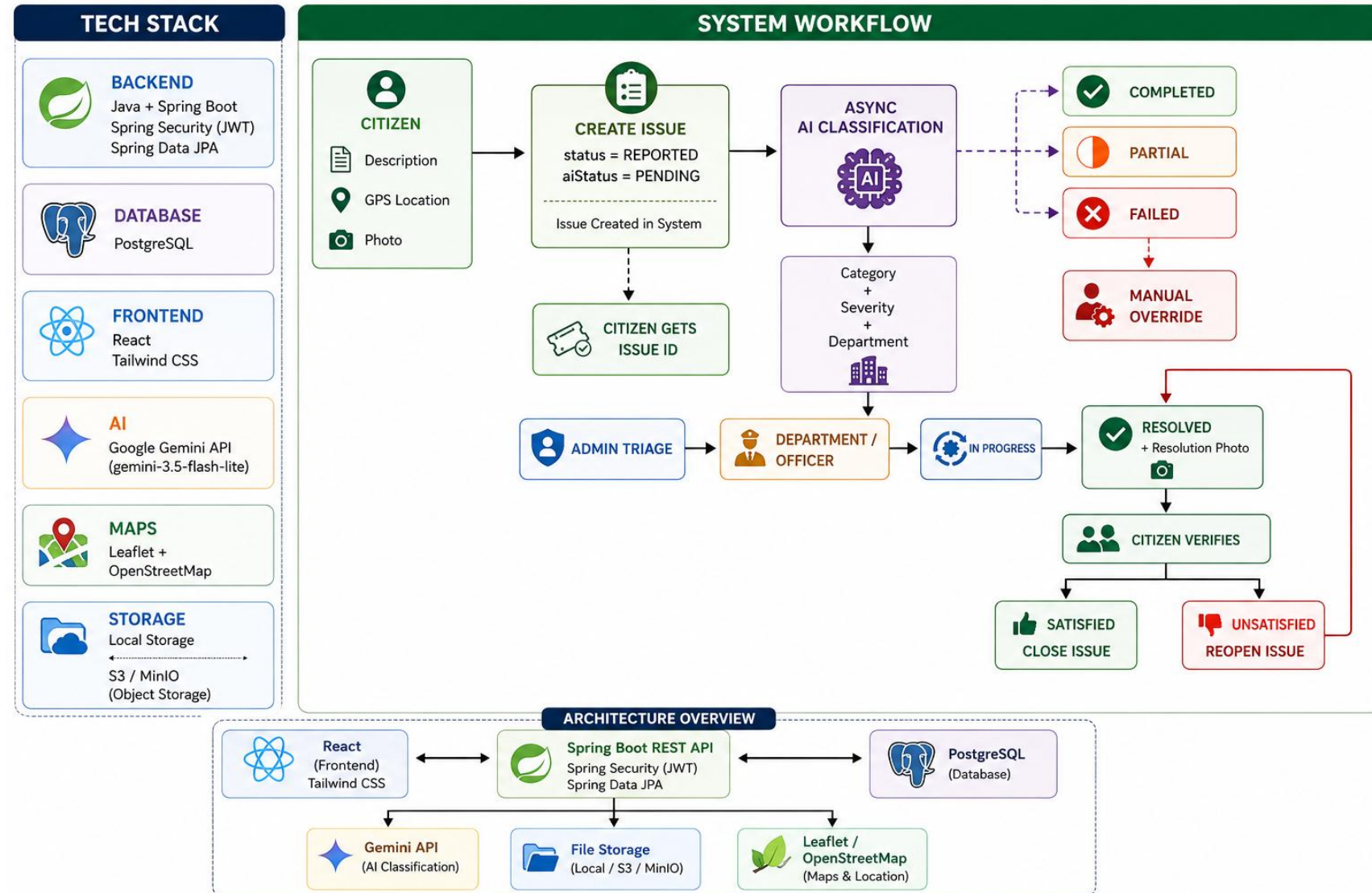
Database- PostgreSQL

Frontend- React, tailwind CSS

Maps- Leaflet + Openstreetmap

AI- Google Gemini API (gemini-3.5-flash-lite)

File storage- Local disk(dev)-> Object storage(S2/MinIO) at scale



Feasibility	Challenges and risks	Strategies for overcoming these challenges
Technical Feasibility • Proven technology stack: Java, Spring Boot, React, PostgreSQL, Gemini & Leaflet/OSM. • Easy integration with municipal systems through REST APIs. Operational Feasibility • Simple reporting using description, GPS and photo. • Supports citizens, officers through role-based workflows. Economic Feasibility ▪ Cost-effective through open-source technologies and scalable storage. ▪ Reduces workload through AI-assisted classification and routing.	1. AI Accuracy Challenge: Poor images/descriptions may cause misclassification. Risk: Incorrect severity or department assignment. 2. False & Duplicate Reports Challenge: Spam, fake and repeated complaints. Risk: Increased workload and field visits. 3. Jharkhand languages Challenge: Limited support for Santali, Mundari, Ho and Kurukh may affect accurate reporting and classification. Risk: Translation or classification errors could misroute complaints or exclude tribal communities.	Strategies for Overcoming These Challenges • AI Validation: Confidence thresholds + human verification. • Spam Control: Spam filtering + duplicate detection. • Data Security: JWT authentication + role-based access + protected storage. • Graceful Failure: Async AI processing with fallback mechanisms. • Keep the workflow language-independent and integrate Adi Vaani as a future tribal-language translation layer.

Potential Impact on People :

Safer & Cleaner Communities: Faster reporting and resolution of potholes, garbage, open drains, broken streetlights and other hazards improves everyday public safety.

Greater Citizen Participation: Gives people a simple way to report local problems and actively contribute to improving their neighbourhood.

Reduced Frustration & Delays: Real-time complaint tracking reduces uncertainty and helps citizens know when and how their issue is being addressed.

Improved Quality of Life: Timely resolution of civic problems creates cleaner, safer, more accessible and better-maintained public spaces.

Benefits of CiviConnect :

Faster Civic Issue Resolution: AI-based classification, prioritization and smart routing help authorities act on issues more efficiently.

Greater Transparency & Accountability: Citizens can track complaints, view progress and verify the final resolution.

Efficient Use of Public Resources: Authorities can prioritize critical issues and assign field workers based on severity and location.

Data-Driven Urban Governance: Civic issue trends and hotspots help authorities identify recurring problems and make better infrastructure and maintenance decisions.

- CitySolution: A complaining task distributive mobile application for smart city corporation using deep learning <https://arxiv.org/abs/2410.12882>
- A Machine Learning Integrated Framework for Automated Civic Complaint Validation and Prioritization <https://doi.org/10.1109/ICSSAS68835.2026.11559493>
- Smart Civic Issue Reporting System with Hybrid AI Classification and Geospatial Analytics for Efficient Urban Governance <https://www.ijert.org/smart-civic-issue-reporting-system-with-hybrid-ai-classification-and-geospatial-analytics-for-efficient-urban-governance-ijertv15is052012>
- CivicAI Complaint Management System <https://ijarcce.com/papers/civica-complaint-management-system>
- CivicFix — Crowdsourced Civic Issue Reporting and Resolution System (SIH25031, Government of Jharkhand) <https://www.ijarsct.co.in/Paper35343.pdf>
- Source of Graph- <https://morth.gov.in/#/>